Group 18: Feature Engineering for Time-Series

Hybrid, Automated, and Interpretable Approaches

CSCE 5222 Feature Engineering | Group 18

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Outline

Part 1 — Presentation

- Recap of Project 1 Proposal
- Datasets and Feature Engineering Techniques
- Machine Learning Models and Baselines
- Results: Classification, Forecasting, Anomaly Detection
- Discussion and Conclusion

- **Part 2 — Live Demo (Google Colab Notebook)**
  - Walkthrough of the Catch22 implementation
  - Live SHAP-guided pruning across all three tasks

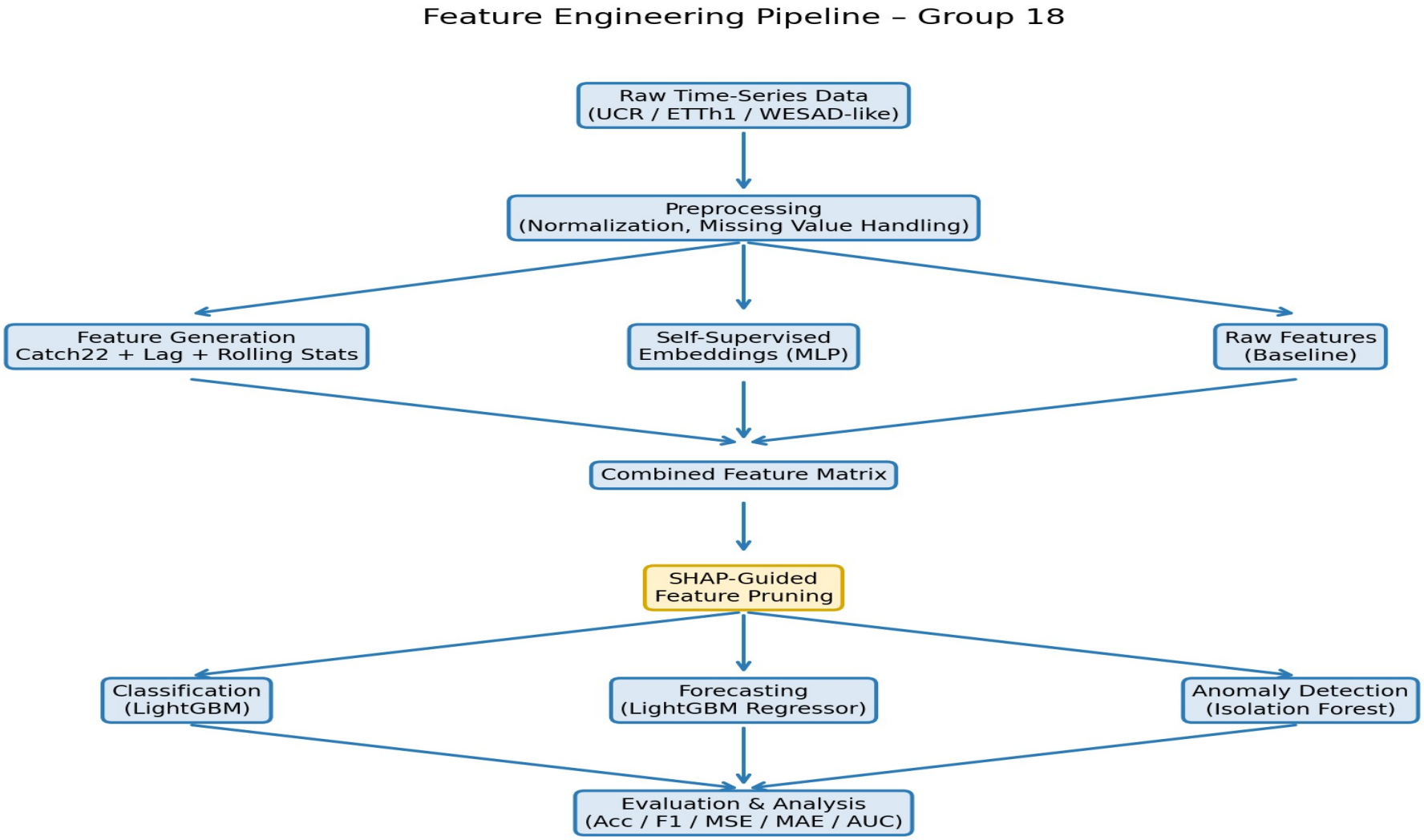
Recap of Project 1 Proposal

- Problem: Extracting meaningful features from time-series is a bottleneck.
- Motivation: Deep learning is powerful but slow and hard to interpret.
- We surveyed 5 papers (2024–2025): COCALITE, SPLiT, PPG Anomaly, Lag, AutoFE-XAI.
- Three datasets: UCR (classification), ETTh1 (forecasting), WESAD-like (anomaly).
- **Phase 1 Feedback Addressed:**
  - Group Number 18 added to all submissions
  - Evaluations clearly separated by task; explicit baselines defined
  - Methodology flowchart completed (was incomplete in Phase 1)

Datasets and Feature Engineering

- **Datasets**
  - UCR (classification): GunPoint, ECG200, ItalyPowerDemand, SyntheticControl, TwoLeadECG
  - ETTh1 (forecasting): 17,420 hourly readings, 7 sensor variables, target = Oil Temperature
  - WESAD-like PPG (anomaly): 2,000-sample BVP signal with injected anomalies (15.9% rate)
- **Four Feature Engineering Strategies**
  - Catch22 — 22 canonical features implemented from scratch in pure NumPy
  - Lag + Rolling Statistics — lags 1 to 24 + rolling mean/std/min/max over 4 windows
  - Self-Supervised MLP Embeddings — 32-unit MLP predicts x[t] from x[t-24:t]
  - SHAP-Guided Pruning — keep top 55–60% of features by mean |SHAP| value

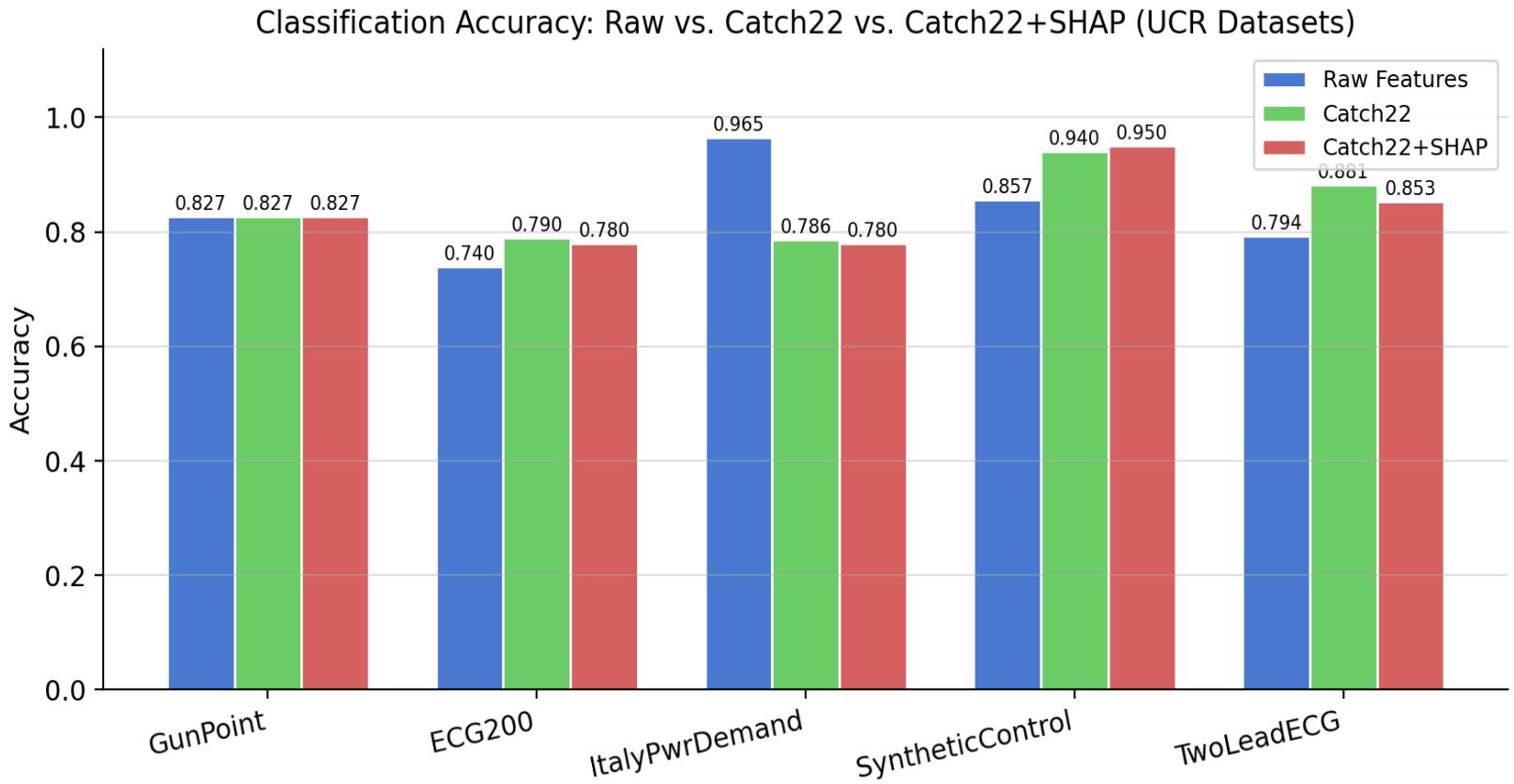
Methodology Pipeline



Machine Learning Models and Baselines

- **Classification**
  - Baseline: GradientBoosting on raw time-series (first 50 steps)
  - Proposed: GradientBoosting on Catch22 → SHAP-pruned | Metrics: Accuracy, F1
- **Forecasting**
  - Baseline: Ridge Regression on lag-1 only
  - Proposed: GradientBoosting on Lag+Rolling+Embeddings → SHAP pruned | MSE, MAE
- **Anomaly Detection**
  - Baseline: Isolation Forest on raw 50-sample windows
  - Proposed: Isolation Forest on Catch22 → SHAP-selected | F1, AUC

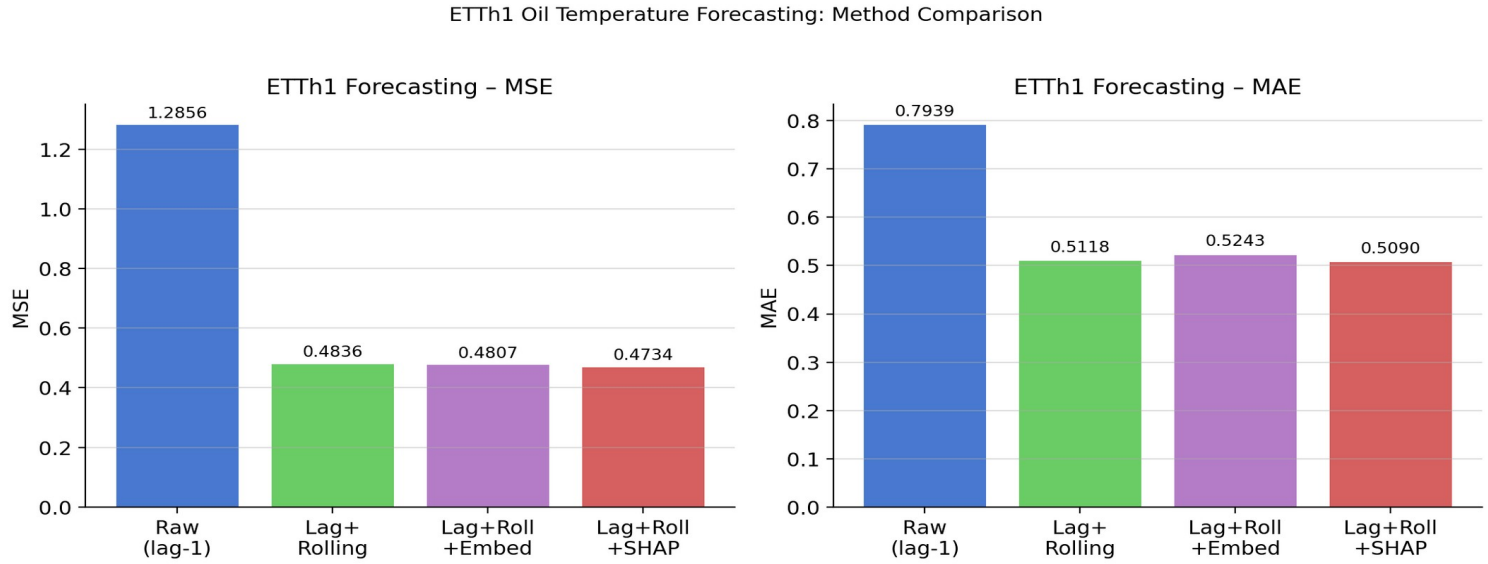
Results: Classification (UCR Datasets)

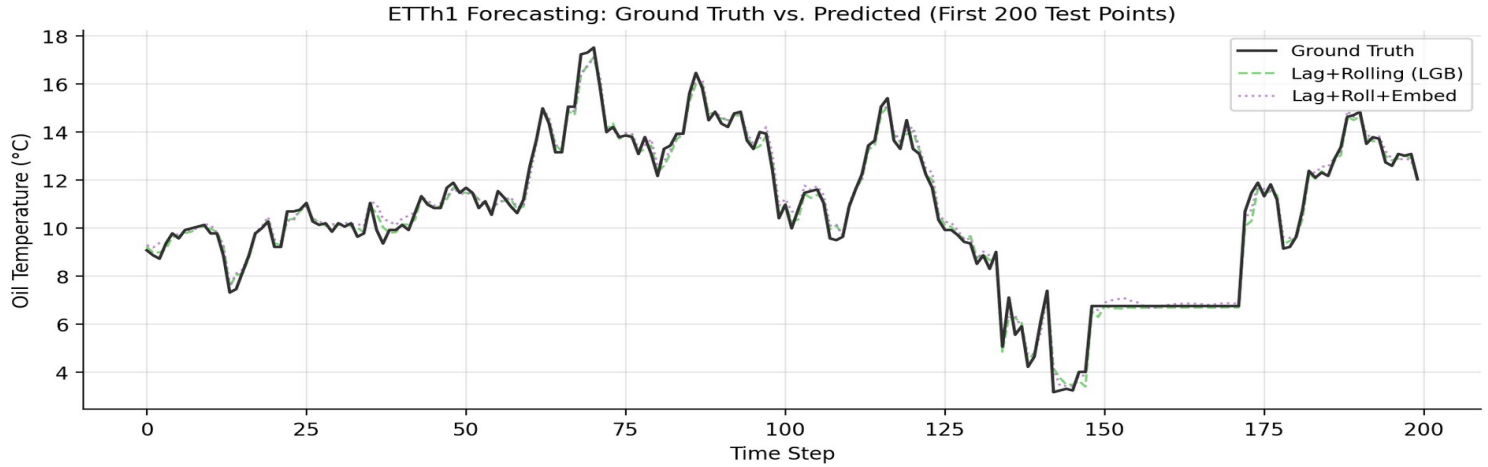


Key Findings:

- **SyntheticControl: SHAP +9.3%**
- **TwoLeadECG: Catch22 +8.8%**
- **ECG200: Catch22 +5.0%**
- ItalyPowerDemand: Raw wins
- (local timing event missed
- by global statistics)
- SHAP pruning: 9/22 removed,
- zero accuracy loss

Results: Forecasting (ETTh1 — Oil Temperature)





Key Results:

Raw (lag-1): MSE = 1.2856

Lag+Rolling: MSE = 0.4836

(−62%)

+ Embeddings: MSE = 0.4807

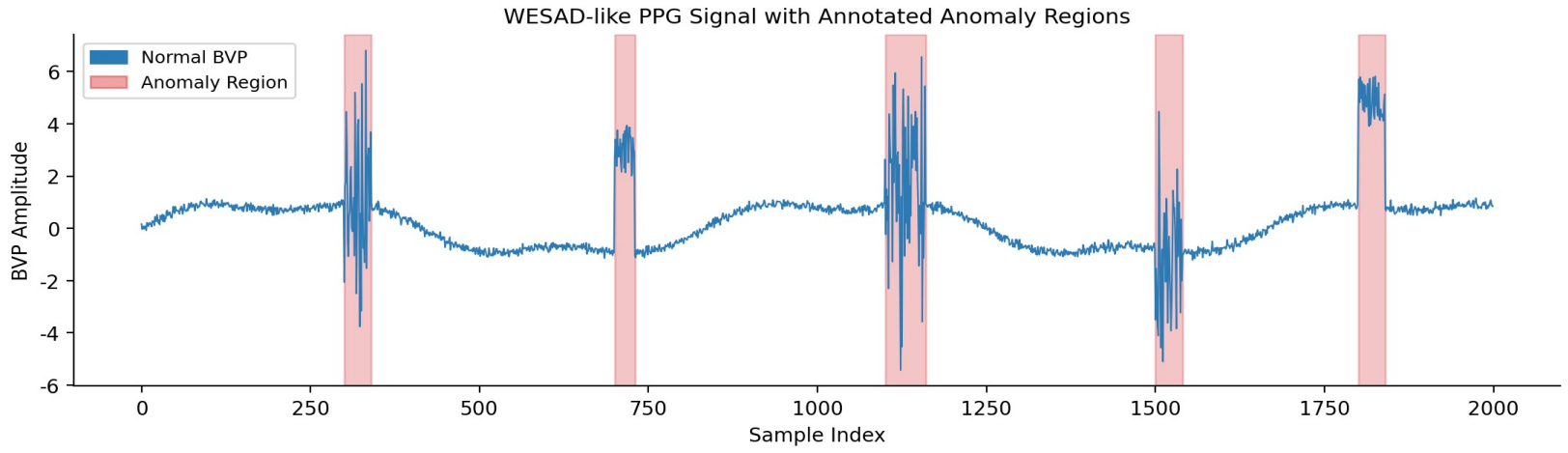
+ SHAP Prune: MSE = 0.4734

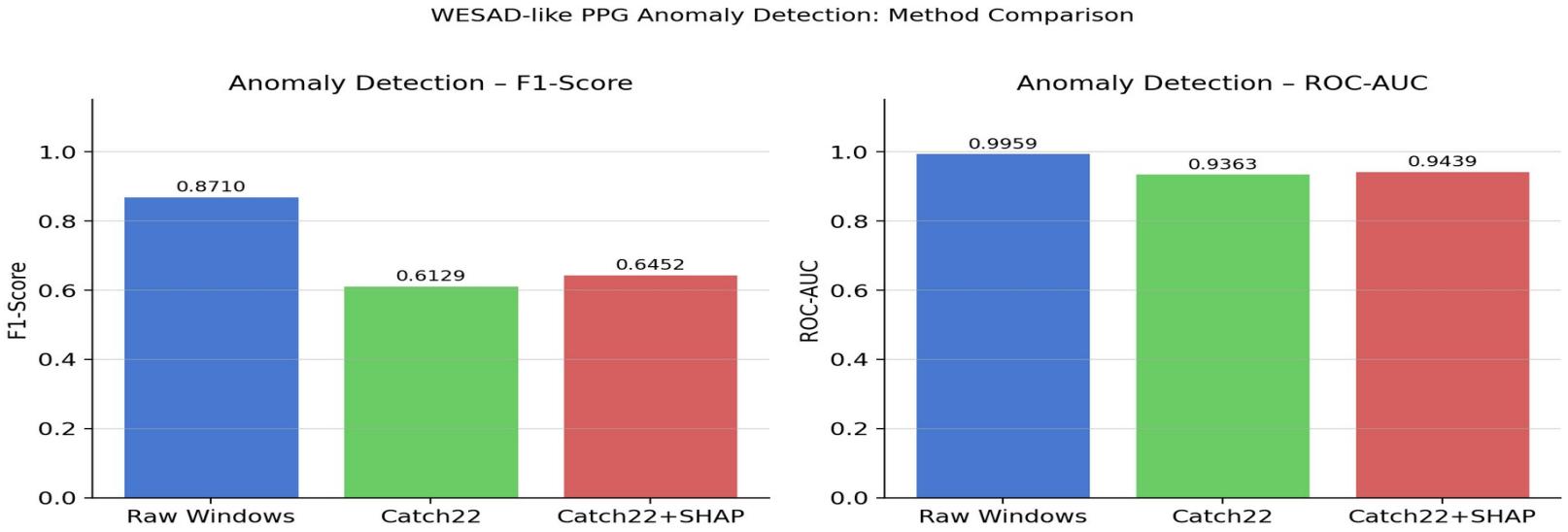
(best)

SHAP removed 12/28 features

Embeddings did not help

Results: Anomaly Detection (WESAD-like PPG)





Key Results:

Raw Windows: F1 = 0.871

Catch22 Only: F1 = 0.613 ↓

Catch22+SHAP: F1 = 0.645 ↑

Why Catch22 hurt:

Normalization removes amplitude information

Why SHAP helped:

Kept diff_variance — captures roughness

Discussion

- **When Catch22 Works Best**
  - Datasets where global statistics differ between classes (SyntheticControl, ECG200, TwoLeadECG)
  - Compresses 150+ time steps into 22 interpretable features; acts as regularization
- **When Raw Features Win**
  - ItalyPowerDemand: class boundary is a sharp drop at one specific hour
  - Catch22 averages over the whole signal, missing local timing events
- **SHAP as a Design Tool**
  - Removed 40–45% of features across all three tasks with zero accuracy loss
  - Faster inference, smaller model — important for edge deployment
- Limitation: WESAD-like data is simulated; real wearable data may differ

Conclusion and Future Work

- **Key Conclusions**
  - Catch22 + GradientBoosting beats raw features on 3/5 UCR datasets (best: +9.3% SyntheticControl)
  - Lag + Rolling Statistics reduce forecasting MSE by 62%; SHAP pruning achieves best MSE of 0.4734
  - SHAP pruning removes 40–45% of features across all tasks with zero accuracy loss
  - Catch22 fails when class differences are local timing events, not global patterns
- **Future Work**
  - Build a lighter Catch22 variant with 10–15 features focused on what SHAP keeps most
  - Validate the anomaly detection pipeline on the real WESAD wearable dataset
  - Explore LLM-assisted feature suggestion based on dataset metadata

References

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